

FTC Autonomous Part 12 Make the L Path Repeatable

Middle and high school FTC students | 80 to 90 minutes | Two motor goBILDA StarterBot

Students repeat the Part 11 autonomous L path, measure where the robot actually stops, and make one justified adjustment. This prepares them to evaluate vision assistance against a reliable, measured baseline.

Lesson outcome

Students can distinguish accuracy from repeatability, document three comparable trials, identify a likely cause of error, and evaluate whether a single change improved the route.

Learning goals

- State the programmed route: 36 inches forward at 0 degrees, turn right to -90 degrees, then 24 inches forward at -90 degrees.
- Measure final forward position, rightward position, final heading, and endpoint distance from the intended finish.
- Distinguish consistent offset from run-to-run variation and use evidence before changing constants.
- Run the complete POWERING ON call-and-clear procedure before every enabled test.

Materials and preparation

- Part 11 two motor robot and StarterBotLPathAutonomous.java; charged battery and Driver Station.
- Open floor area of at least 8 by 8 feet, painter tape, measuring tape, right angle square, printed record sheet, and POWERING ON poster.
- Confirm left_drive, right_drive, and imu names; motor direction; encoder counts per inch; Control Hub mounting and IMU orientation; wheel and cable security.
- Tape the start, nominal finish, an L path with side buffer, and a clear stopping area. Mark one fixed point on the robot chassis as the measurement reference.

Coordinate rule: x is forward from the start, y is to the robot's right at the start. The target endpoint is (36, 24) inches. Right-turn heading is -90 degrees for the Part 11 setup. Confirm the turn sign on the real robot before testing.

Route boundary and POWERING ON

Use the same route, start orientation, robot reference point, power settings, battery condition, and floor surface for comparable runs. The complete travel area, turn sweep, side buffer, and stopping buffer remain empty while the robot is enabled.

1. Set the robot at the start while the Driver Station shows DISABLED. Complete any hands-on adjustment now.
2. Remove measuring tools, hands, loose cables, and students from the complete route.
3. Safety captain looks around and asks, "Route clear?"
4. Safety captain loudly calls, "POWERING ON!"
5. Pause so everyone can step behind the boundary; visually confirm the whole route is clear.
6. Safety captain authorizes INIT and START. Operator remains ready to press STOP.
7. After motion ends, operator presses STOP and visibly confirms DISABLED. Only then may the safety captain open the route for measurement.

Stop a trial immediately when

- A person, cable, tool, or obstruction enters the movement boundary.
- The robot leaves the course, heads the wrong way, oscillates, or continues beyond its expected stop or timeout.
- Encoder or yaw telemetry is absent, frozen, or implausible; a wheel, battery, or Hub appears loose.

Student roles

Role	Action
Safety captain	Checks route and authorizes each enabled trial.
Operator	Runs the OpMode and presses STOP after each attempt.
Course marshal	Squares the disabled robot at the start and clears tools.
Measurers	Enter only after DISABLED; measure x, y, and heading.
Data recorder	Records run conditions, exit reason, and measured results.
Debugging analyst	Proposes exactly one evidence-supported change.

Lesson sequence

Stage	Minutes	Teacher move and student evidence
Predict	8	Review Part 11. Mark (36, 24) and -90 degrees; ask whether hitting one endpoint once establishes reliability.
Model measurement	10	Demonstrate the chassis reference point, x and y axes, and measuring only after DISABLED.
Readiness and roles	10	Inspect mechanics and sensors; assign safety captain, operator, marshal, measurers, recorder, and analyst.
Baseline trials	22	Run the full safety call-and-clear for each of three trials. Record endpoint and exit reason before resetting.
Analyze	10	Compare offsets and variation; ask which evidence indicates a consistent error or an intermittent problem.
One change	8	Record a prediction and modify one approved code constant or mechanical item. Keep every other condition stable.
Retest	16	Repeat two or three trials with full safety sequence; compare with baseline.
Exit response	6	Students explain improvement, remaining error, and why a camera would need a reliable driving baseline.

Part 12 video pause guide

These pause times follow the 2-minute whiteboard production plan. Adjust them to the actual video after the final CapCut edit.

Pause	Teacher question	Expected idea
0:27	What must stay the same across trials?	Starting point, heading, robot reference point, surface, and autonomous settings.
0:36	Can one perfect finish establish repeatability?	No; accuracy and run-to-run variation require multiple observations.
0:54	Which parts of the course must be clear?	Both legs, full turn sweep, side boundaries, and stopping buffer.
1:05	When may the operator press START?	After POWERING ON, a pause, everyone behind the line, and visual route check.
1:25	When may students enter the course?	Only after STOP, visible DISABLED, and the safety captain opening the route.
1:45	What pattern suggests a calibration test?	A repeatable distance miss; test straight travel separately before changing counts per inch.

Hands on measurement walkthrough

Before the first run, place the disabled robot with the same marked chassis point over the taped origin and square its nose to the x axis. The measurers point to the target finish without entering the course during an enabled trial. The data recorder writes down the current constants, floor surface, and battery state.

The safety captain performs the full call-and-clear. The operator starts the Part 11 OpMode and watches for any need to press STOP. Nobody follows the moving robot. When the motion ends, the operator presses STOP and confirms DISABLED. The safety captain opens the route. Students then measure x and y of the marked chassis point relative to the start and note final yaw or a floor-measured heading. They record whether the OpMode completed normally or exited by timeout or STOP.

Reset the robot to the same start orientation and repeat until there are three baseline trials. A trial that times out is recorded as a failed trial, not quietly discarded. If the robot makes unsafe or unpredictable movements, stop the activity and repair the cause before continuing.

Reading the results

Endpoint error for each run is $\sqrt{(x - 36)^2 + (y - 24)^2}$, in inches. The signed coordinate errors are $x - 36$ and $y - 24$. Heading error is the shortest signed angle from -90 degrees to the measured final yaw. Students may use a calculator; first ask them to explain each subtraction in words.

Observation	First check
Similar endpoint offset each run	Review calibration and whether the first leg, turn, or second leg contributes.
Large run-to-run spread	Inspect starting alignment, loose parts, wheel slip, inconsistent battery or floor contact.
Distance misses, heading near target	Test a straight leg separately before recalibrating counts per inch.
Final angle misses consistently	Check IMU orientation, turn target, tuning, and reported exit reason.
Timeout or stalled wheel	Stop testing; diagnose friction, wiring, motor mode, and timeout before retesting.

If a separate straight-line test confirms a consistent distance scale error, an estimate is $\text{new counts per inch} = \text{old counts per inch} \times \text{commanded distance} \div \text{measured distance}$. Do not apply that estimate to the L-path endpoint alone: both straight legs and the turn contribute to its final x and y.

Success evidence

- Three baseline trials and at least two retests recorded with comparable start conditions.
- Every measurement happens only after DISABLED is confirmed.
- Students identify whether the change improved average endpoint error and whether the results remain consistent.
- Students can distinguish target-relative camera information, introduced next, from measured robot motion.

Printable student trial record

Team: _____ Date: _____ Robot reference point: _____

Target endpoint: x = 36 in forward, y = 24 in right, heading = -90 degrees. Start at (0, 0), heading 0 degrees.
Record the same chassis point after each run.

Baseline settings: counts/in _____ drive power _____ floor/battery _____

Trial	x in	y in	Final yaw	Endpoint error in	Exit reason / observation
Baseline 1	_____	_____	_____	_____	_____ _____
Baseline 2	_____	_____	_____	_____	_____ _____
Baseline 3	_____	_____	_____	_____	_____ _____

Baseline pattern: _____

One change (and why): _____

Prediction: _____

Trial	x in	y in	Final yaw	Endpoint error in	Exit reason / observation
Retest 1	_____	_____	_____	_____	_____ _____
Retest 2	_____	_____	_____	_____	_____ _____
Retest 3	_____	_____	_____	_____	_____ _____

Mean baseline endpoint error: _____ in Mean retest endpoint error: _____ in

Did the robot finish closer to the target more consistently? What supports your answer?

What additional test would separate a distance calibration problem from a turn problem?

Safety captain initials for each completed call-and-clear: B1 ____ B2 ____ B3 ____ R1 ____ R2 ____ R3 ____